

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : PHAS0042

ASSESSMENT : PHAS0042A6UC
PATTERN

MODULE NAME : PHAS0042 - Quantum Mechanics

LEVEL: : Undergraduate

DATE : 11/05/2021

TIME : 14:30

This paper is suitable for candidates who attended classes for this module in all years

Additional material	N/A
Special instructions	N/A
Exam paper word count	N/A

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Answer ALL questions

Estimated time to complete paper: 3 hours and 30 minutes

The numbers in square brackets in the right-hand margin indicate the provisional allocation of maximum marks per question section.

1. (a) The Hamiltonian $\hat{H}$ of a quantum harmonic oscillator subject to a perturbation of arbitrary strength is given by:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 + g_1\hat{x}\hat{p} + g_2\hat{p}\hat{x}$$

where g_1 and g_2 are constants that can be complex. Re-express $\hat{H}$ in terms of raising and lowering operators, for the case $g_1 = g_2$, explaining your notation.

[5]

- (b) Considering a basis of states with even quanta $n = 0, 2$ only, give $\hat{H}$ in matrix form, for the case that $g_2 = 0$. Is the resulting matrix Hermitian?

[5]

- (c) Obtain the eigenvalues of $\hat{H}$ for the case where $g_1 = i\omega$ and $g_2 = 0$.

[5]

- (d) Using the full basis, the system is prepared in a state $|\Psi_\alpha\rangle$ that is an eigenfunction of the lowering operator with eigenvalue α , thus $\hat{a}_-|\Psi_\alpha\rangle = \alpha|\Psi_\alpha\rangle$, corresponding to a superposition:

$$|\Psi_\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

From the above expression, or otherwise, show that $\langle\Psi_\alpha|\hat{a}_+|\Psi_\alpha\rangle = \alpha^*$.

[2]

Hence, obtain the expectation value of the momentum operator $\hat{p}$ in this state.

[3]

2. A quantum system evolves under the action of a Hamiltonian $\hat{H}$ with a complete set of eigenstates $|\chi_{\pm}\rangle$ and corresponding eigenvalues $E_{\pm} = \pm\epsilon$. An arbitrary quantum state $|\Psi\rangle$ may be represented either in this two-state basis $|\Psi\rangle = d_+|\chi_+\rangle + d_-|\chi_-\rangle$ or using an alternative representation $|\Psi\rangle = \sum_{j=1,2} b_j|\psi_j\rangle$ using a complete basis of orthonormal basis states. A change of representation may be achieved by the action of a matrix on two vectors $\mathbf{d} = \mathbf{S} \mathbf{b}$.

- (a) Consider the case where the matrix $\mathbf{S}$ is given by:

$$\mathbf{S} = \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix}$$

If the eigenvectors of $\hat{H}$ are given in the $|\psi_k\rangle$ basis as $|\chi_{\pm}\rangle = \sum_k C_k^{\pm}|\psi_k\rangle$, obtain the coefficients $C_k^{\pm}$ in terms of β . Explain any possible ambiguities and whether there might be more than one choice. For your answer, assume that in the eigenvector basis $\hat{H}_{11} = \langle\chi_-|\hat{H}|\chi_-\rangle$. [5]

- (b) The system considered above is initially prepared in state $|\Psi(t=0)\rangle = i|\psi_2\rangle$ at a time $t = 0$. What vector $\mathbf{b}$ corresponds to this state? For the case $\beta = \pi/4$, write down a unitary operator $\hat{U}(t, 0)$ in matrix form that will evolve the state to a later time t . Hence obtain $\mathbf{b}(t) = \hat{U}(t, 0)\mathbf{b}(t=0)$. [5]
- (c) Hence or otherwise give the probability of measuring the system in state $|\Psi\rangle = |\psi_1\rangle$ as a function of time. [2]

- (d) An example of a two-state space is a spin $1/2$ particle. Two particles have spin angular momenta $s_1 = 1/2$ and $s_2 = 1/2$ respectively. They experience a Hamiltonian:

$$\hat{H} = C_{12}\mathbf{S}_1 \cdot \mathbf{S}_2 + \beta_0(\hat{S}_{1z} + \hat{S}_{2z})$$

which includes an external magnetic field of magnitude β_0 as well as a mutual interaction of strength C_{12} . Express the interaction term in $\mathbf{S}_1 \cdot \mathbf{S}_2$ using raising and lowering operators $\hat{S}_{\pm 1}$ and $\hat{S}_{\pm 2}$. [3]

- (e) Hence represent the above Hamiltonian as a matrix, being careful to define all terms and obtain its eigenvalues and eigenvectors. [5]

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3. A particle of mass m moves in a one-dimensional infinite potential well defined by $V(x) = 0$ when $|x| \leq a$, $V(x) = \infty$ otherwise. The energy eigenvalues are:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2} = Cn^2, \quad \text{with } n = 1, 2, 3, \dots$$

The corresponding orthonormal eigenfunctions are the even and odd functions

$$\psi_n(x) = \begin{cases} \frac{1}{\sqrt{a}} \cos(nbx) & \text{for } n = 1, 3, 5 \dots \\ \frac{1}{\sqrt{a}} \sin(nbx) & \text{for } n = 2, 4, 6 \dots \end{cases}.$$

with $b = \frac{\pi}{2a}$.

The potential between $-a < x < a$ is modified to:

$$V(x) = \lambda C \sin(3bx)$$

(a) Using Perturbation Theory:

- i. Determine first-order order correction to the energy of the ground state. [3]
- ii. Determine second-order correction of the ground state. [6]
- iii. For what values λ would you expect perturbation theory to be useful? [2]

(b) Use the variational principle to obtain an energy for the ground state using the trial function

$$\phi_T(x) = \frac{1}{\sqrt{a}} \cos(bx) + q \frac{1}{\sqrt{a}} \sin(2bx) \equiv \psi_1(x) + q\psi_2(x)$$

where q is a variational parameter. [5]

For the case where $\lambda = \frac{1}{2}$ find the best value for q and hence the optimal energy. [2]

Suggest an improved trial function which might further lower the energy. [2]